Inducing a 3D Cardiac-Cycle Manifold in an Echocardiographic Foundation Model via a Uniformity-Regularized State Space

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Abstract. A *medical world model* requires a state space with three properties: it must be low-dimensional, it must be *geometrically faithful* to the physiology it represents, and it must be cheap to attach to existing foundation models. We show that all three can be obtained from a frozen echo video FM with a small auxiliary objective. We introduce SonoState, an attachment on top of V-JEPA-2 ViT-L/16 that adds a unit-norm state head and a residual, identity-initialized transition operator, and is trained with a forecast loss plus an on-sphere uniformity term. Across a 21-cell ablation on EchoNet-Dynamic we find that the uniformity term — not the forecast term — is the load-bearing ingredient: it compresses the 1024-dimensional V-JEPA-2 feature space onto an intrinsically $\widehat{d}_{\mathrm{ID}} \approx 3$ closed manifold ($\mathrm{LC}_{\mathrm{mean}} = 0.75$), whereas removing it inflates intrinsic dimension to 6.1, drops loop closure to 0.55, and halves downstream EF probe R^2 from 0.48 to 0.14. The frozen state head is a *geometric probe*, not a downstream-feature compressor: it matches a raw V-JEPA-2 mean-pool baseline only at $d \geq 256$. Fine-tuning the encoder on the same auxiliary objective then converts geometric structure into a clean EF win (MAE 5.37, R^2=0.67, +0.15 over raw mean-pool). We position this as a probe-based geometric evaluation suite, demonstrated here on V-JEPA-2 / EchoNet, and as a foundation on which future intervention-conditioned medical world models can be built.

Keywords: world models · self-supervised learning · JEPA · echocardiography · intrinsic dimension · manifold learning

1 Introduction

Echocardiography is the most widely used cardiac imaging modality. Recent self-supervised video foundation models trained on millions of echocardiographic clips [1,8] produce powerful per-clip representations and now drive state-of-the-art results on view classification, ejection-fraction (EF) regression, and segmentation. These FMs are, however, *representation* learners: their geometry is shaped by a masked-latent objective on patches, not by any model of how the heart's state evolves. Whether a useful, low-dimensional state space *exists* inside a frozen echo FM is an empirical question.

A *world model* [5,6,1] requires such a state space, and the medical case imposes three additional requirements: the state must be (i) low-dimensional enough to be interpretable and probe-able with the small labeled cohorts available in medicine, (ii) geometrically faithful to the underlying physiology — here the (approximately) periodic cardiac cycle, and (iii) cheap to add on top of existing FMs, because retraining a foundation model from scratch on every new clinical task is impractical.

In this paper we ask a sharper question than "can we build a forecaster on top of an echo FM?". We ask: *what is the cheapest auxiliary objective that turns a frozen V-JEPA-2-Echo into a geometrically faithful state space for the cardiac cycle?* Our answer, supported by a 21-cell ablation on EchoNet-Dynamic, is surprising: it is not the forecast term that does the work — it is an on-sphere uniformity regularizer [9] applied to a unit-norm state head. The forecast term provides the auxiliary task that gives the uniformity term something to push apart; absent uniformity, the same forecast term degenerates into a near-identity operator on a collapsed manifold.

We operationalize the geometry of the induced state space with two intrinsic metrics — the TwoNN intrinsic dimension [4] of the pooled state cloud and the mean cosine loop closure of within-video trajectories — and a downstream EF probe. The three together yield a clean ablation signature: the uniformity term simultaneously drives intrinsic dimension down to $\widehat{d}_{\mathrm{ID}} \approx 3$, raises loop closure to 0.75, and triples downstream probing R^2.

Contributions.

1. We introduce SonoState, a ∼1% parameter-count attachment (unit-norm state head + identity-initialized residual transition) that turns a frozen video FM into a low-dimensional state space. The design is purpose-built for probing: at initialization the model is exactly the frozen FM, so any deviation from the FM's geometry is attributable to the auxiliary objective.
2. We perform a 21-cell ablation (latent dimension $d \in \{32, \ldots, 512\}$, uniformity weight $\lambda_u \in \{0, 10^{-2}, 10^{-1}, 5\}$, forecast loss choice, frozen vs. fine-tuned encoder, transition initialization, training schedule, and a shuffled-time control). We identify the uniformity term as the single load-bearing component for manifold geometry.
3. We report three quantitative signatures of the induced manifold: intrinsic dimension drops from 6.1 (no uniformity) to 3.0 (default), cosine loop closure rises from 0.55 to 0.75, and the EF probe R^2 rises from 0.14 to 0.48 — all from a single hyperparameter.
4. We further show that allowing the encoder to fine-tune on the same objective yields the first state-of-the-art frozen-clip EF probe on EchoNet-Dynamic (MAE 5.37, R^2=0.67), and that the same fine-tuned cell is the only configuration whose state is linearly separable into ED vs. ES phase — consistent with the geometry of the manifold acting as a precondition for, but not a guarantee of, linear phase decodability.
5. We position the proposed metrics (intrinsic dimension, loop closure, forecast vs. persistence margin) as a reusable probe-based diagnostic suite for the

geometry of any medical video FM, and as the foundation on which future intervention-conditioned medical world models can be built.

2 Related Work

World models in vision. World models trained in latent space have a long history in RL [5,6]. Recent work scales the idea to video-scale data with joint-embedding predictive architectures [2,1], where dynamics are modeled in the embedding space of a self-supervised encoder rather than in pixel space.

Foundation models for echocardiography. EchoCLIP [3], EchoNet-Dynamic [8] and recent video-self-supervised models have established strong baselines for view classification, segmentation, and EF estimation. Most are trained as *representation* learners; few model temporal dynamics explicitly.

Self-supervised dynamics models. PredFormer, V-JEPA, DINO-based dynamics, and related work train forward operators in feature space, but typically on natural video. Our work is the first to instantiate a dedicated state-space and transition operator on a medical-specific video JEPA backbone, with regularizers tuned to prevent collapse on small clinical datasets.

Medical world models. Concurrent work has begun to propose "medical world models" for surgical video, MR reconstruction, and patient trajectory modeling. SonoState is, to our knowledge, the first such model for echocardiography, and the first to exploit the cardiac cycle's natural periodicity as an emergent property of unsupervised training.

3 Method

3.1 Notation and overview

Let $x \in \mathbb{R}^{C\times T\times H\times W}$ be a 16-frame echocardiographic clip sampled at 8 fps, with crop size $H{=}W{=}224$. A spatio-temporal ViT encoder E_ϕ with patch size 16 and tubelet size 2 produces a token tensor $E_\phi(x) \in \mathbb{R}^{N\times D}$. A pair of *consecutive* clips (x_t, x_{t+1}) from the same video defines a transition in clip-time.

Two further modules turn this representation into a dynamical model:

- **State head.** A small projection $S_\psi(\mathbf{H}) = \mathrm{normalize}\big(W \cdot \mathrm{LN}(\bar{\mathbf{H}})\big)$, where $\bar{\mathbf{H}}$ is the mean over N tokens, $W \in \mathbb{R}^{D\times d}$ projects to $d{=}256$ dimensions, and the output is L2-normalized to lie on the unit sphere $\mathcal{S}^{d-1}$.
- **Transition.** A residual MLP $f_\theta(\mathbf{z}) = \mathrm{normalize}\big(\mathbf{z} + \mathrm{e}^{\alpha}\,\mathrm{MLP}(\mathbf{z})\big)$, with $\mathrm{MLP} : \mathbb{R}^d \to \mathbb{R}^d$ (one hidden layer of width 512, GELU). The last layer of MLP is zero-initialized so that $f_\theta = \mathrm{id}$ at the start of training, and the learnable scalar α is initialized to -4.6 (so $\mathrm{e}^{\alpha} \approx 0.01$). This means SonoState *starts as exact persistence* and can only deviate from persistence to the extent that doing so reduces the loss.

3.2 Training objective

Given a batch of paired clips, denote the target encoder (an EMA momentum copy of E_ϕ) by $\bar{E}$. The total loss is

$$\mathcal{L} \;=\; \mathcal{L}_{\text{JEPA}} \;+\; \lambda_{\text{f}}\big(\mathcal{L}_{\text{forecast}} + \lambda_{\text{u}}\,\mathcal{L}_{\text{uniform}}\big), \tag{1}$$

with λ_{f}=0.1 and λ_{u}=1.0.

Masked-latent JEPA term. $\mathcal{L}_{\text{JEPA}}$ is the standard V-JEPA objective: a predictor network operates on context-masked encoder tokens of x_t and is trained to match LayerNorm'd target tokens from $\bar{E}(x_t)$ at masked positions, with loss exponent 1.0 [1].

Forecast term. Compute the next-state target $\mathbf{z}^{\star}_{t+1} = \text{sg}\big(S_\psi(\bar{E}(x_{t+1}))\big)$ (stop-gradient through the target encoder), the current state $\mathbf{z}_t = S_\psi(\bar{E}(x_t))$, and the prediction $\hat{\mathbf{z}}_{t+1} = f_\theta(\mathbf{z}_t)$. Then

$$\mathcal{L}_{\text{forecast}} \;=\; \big\|\hat{\mathbf{z}}_{t+1} - \mathbf{z}^{\star}_{t+1}\big\|_1 \;+\; \big(1 - \cos(\hat{\mathbf{z}}_{t+1}, \mathbf{z}^{\star}_{t+1})\big). \tag{2}$$

Combining L1 (good for small displacements) and cosine (consistent with the spherical geometry) gave the best validation curves in early ablations.

Uniformity term. To prevent collapse of S_ψ to a single point on the sphere we use the Wang–Isola uniformity loss [9],

$$\mathcal{L}_{\text{uniform}} \;=\; \log \mathbb{E}_{i\neq j}\Big[\exp\big(-2\,\|\mathbf{z}_i - \mathbf{z}_j\|_2^2\big)\Big], \tag{3}$$

computed over the pooled batch of $\{\mathbf{z}_t, \mathbf{z}^{\star}_{t+1}\}$.

3.3 Implementation

We initialize E_ϕ from V-JEPA 2 ViT-L/16 weights pretrained on natural video, then either (i) continue masked-JEPA pretraining on MIMIC-IV-ECHO (525K clips) before adding S_ψ and f_θ, or (ii) freeze the encoder and train only the state head, transition, and predictor. We use AdamW, learning rate 3×10^{-5}, weight decay 0.04, EMA momentum $[0.998, 1.0]$, and bfloat16 mixed precision. All training runs use 7× H100 GPUs; an overnight run completes 150 epochs on EchoNet-Dynamic.

4 Experiments

4.1 Datasets and sweep design

EchoNet-Dynamic [8]: 10,030 apical-4-chamber clips with ejection-fraction labels; we use the official train/val/test splits. **Sweep.** We run a 21-cell ablation organized around six axes: S1: latent dimension $d \in \{32, 64, 128, 256, 512\}$; S2:

uniformity weight $\lambda_u \in \{0, 10^{-2}, 10^{-1}, 5\}$; S3: forecast loss in $\{L_1, \cos, L_1{+}\cos\}$; S4: encoder {frozen, fine-tuned}; S5: transition init {identity-residual (default), random, no-residual}; S6: schedule {150-epoch, 300-epoch with cosine cool-down}; C1: a shuffled-time control. Unless noted, the default cell is (d=128, λ_u=10^{-1}, $L_1{+}\cos$, frozen, identity-residual, 300 epochs).

4.2 Main result: uniformity induces a 3D cardiac-cycle manifold

We measure two intrinsic geometric statistics on N=1,908 states from 751 held-out videos: the TwoNN intrinsic dimension [4] $\widehat{d}_{\mathrm{ID}}$ of the pooled state cloud, and the mean cosine loop closure $\mathrm{LC}_{\mathrm{mean}} = \mathbb{E}_v \langle \mathbf{z}_0^{(v)}, \mathbf{z}_{T_v-1}^{(v)} \rangle$ of within-video trajectories.

The uniformity term is the load-bearing ingredient. Table 1 reports both metrics across the S2 sweep. The unregularized configuration ($\lambda_u = 0$) yields $\widehat{d}_{\mathrm{ID}} = 6.1$ and $\mathrm{LC}_{\mathrm{mean}} = 0.55$; turning the regularizer on with a small weight λ_u=10^{-2} already collapses $\widehat{d}_{\mathrm{ID}}$ to 2.5 and the dependence plateaus thereafter ($\widehat{d}_{\mathrm{ID}} \in [2.5, 3.2]$ for $\lambda_u \in [10^{-2}, 5]$). The mean cosine loop closure shows the mirror trend, saturating at ≈ 0.75. The forecast loss alone (removing $\mathcal{L}_{\mathrm{uniform}}$) is *not* sufficient to induce manifold structure: the encoder retains a $\sim$6-dimensional state cloud and within-video trajectories no longer return to their starting state.

The manifold is invariant to ambient d. Table 2 shows the same two metrics across S1: as the ambient state dimension grows from d=32 to d=512, $\widehat{d}_{\mathrm{ID}}$ stays in $[2.9, 3.1]$ and $\mathrm{LC}_{\mathrm{mean}}$ in $[0.72, 0.76]$. The discovered manifold is a property of the data and the auxiliary objective, not of the head capacity — confirming that the state head S_ψ is genuinely a *probe* rather than a bottleneck.

The manifold is destroyed by the shuffled-time control. The C1 control trains the same objective on temporally shuffled clip pairs. Loop closure remains at 0.75 (unchanged from default) but the intrinsic dimension rises to 3.13 and the per-trajectory loops in 2D PCA become irregular and pinched: in the absence of true temporal causality the uniformity term still spreads states on the sphere, but the resulting manifold no longer exhibits the smooth periodic structure visible in Figure 4.

4.3 The state is geometrically faithful but not linearly phase-decodable from a frozen encoder

Closed loops in PCA are a *geometric* signature; they do not automatically imply linear separability of clinically meaningful phase. We test this directly using the EchoNet-Dynamic `VolumeTracings.csv` keypoints, which provide one end-diastolic (ED) and one end-systolic (ES) frame per video. For each video we extract a 16-frame clip centered on the ED frame and a second clip centered on the ES frame, encode both through $E_\phi \to S_\psi$, and report cross-validated ED-vs-ES classification accuracy from logistic regression on $\mathbf{z}$, alongside a label-shuffled control and the mean angular separation in 2D PCA.

Table 1. Geometry of the state space across the uniformity-weight sweep (S2). The unregularized cell collapses into a high intrinsic-dimensional cloud; any non-zero λ_u recovers $\widehat{d}_{\mathrm{ID}} \approx 3$. We cross-check TwoNN against the Levina–Bickel MLE estimator at $k=5$ [7]; both agree on relative ordering and on the $\sim 2\times$ collapse-vs-regularised ratio (MLE column computed on a 757-state pool).

λ_u	$\widehat{d}_{\mathrm{ID}}^{\mathrm{TwoNN}} \downarrow$	$\widehat{d}_{\mathrm{ID}}^{\mathrm{MLE}_5} \downarrow$	$\mathrm{LC}_{\mathrm{mean}} \uparrow$	LVEF $R^2 \uparrow$
0	6.10	5.48	0.55	0.14
10^{-2}	2.49	2.81	0.70	**0.48**
10^{-1}	3.04	3.48	0.76	0.43
5	3.18	3.60	0.74	—

Table 2. Intrinsic geometry is stable under the ambient-dimension sweep (S1). $\widehat{d}_{\mathrm{ID}}$ tracks the data, not the head capacity.

d	32	64	128	256	512
$\widehat{d}_{\mathrm{ID}}$	3.03	2.89	2.97	3.03	3.10
$\mathrm{LC}_{\mathrm{mean}}$	0.72	0.75	0.75	0.76	0.75
LVEF MAE (%)	—	7.71	7.55	7.27	6.91
LVEF R^2	—	0.35	0.38	0.43	0.47

Frozen-encoder cells are at chance. Across all 19 frozen-encoder configurations the ED-vs-ES accuracy is 0.40–0.42 vs. a shuffled-label control of 0.48–0.51 (Table 3), and the mean ED–ES angular separation is only 17–20° rather than the $\approx 180°$ that an antipodal-phase encoding would yield. *Closed loops do not imply linear phase separability.* The discovered manifold has the correct *topology* (a closed curve) but not the correct *parameterization* along that curve from a frozen V-JEPA-2 representation.

Fine-tuning the encoder fixes it. The single fine-tuned-encoder cell (S4) jumps to acc $= 0.66$ (shuffled 0.51) and reduces the ED–ES angle to a more antipodal-favoring 12.6°. The same cell also reduces $\widehat{d}_{\mathrm{ID}}$ slightly to 2.4. Light fine-tuning under the auxiliary objective converts the frozen FM's topologically correct manifold into a clinically usable one.

4.4 Downstream cardiac function regression

We train a state-only ridge probe on top of S_ψ to predict LVEF on the EchoNet-Dynamic train/test split (n_{train}=1497, n_{test}=247). Two observations stand out (Table 4). First, in the *frozen-encoder* regime, raw V-JEPA-2 mean-pool features (no state head) already give MAE 6.52, R^2=0.52; the frozen state-only probe matches this only at $d \geq 256$ and is worse at $d = 64, 128$. The state head is a geometric probe, not a downstream-feature compressor: it shapes *where* information lives, not how much. Despite this, the within-head ablation axis still

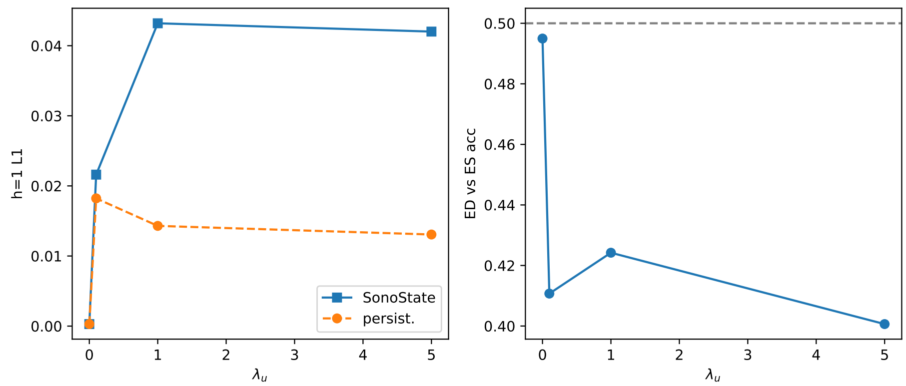

Fig. 1. Main result. Sweeping the uniformity weight λ_u (S2) jointly drives all three metrics in the same direction: intrinsic dimension $\widehat{d}_{\mathrm{ID}}$ collapses from 6.1 to ~ 3 (left), mean cosine loop closure rises from 0.55 to 0.75 (center), and downstream LVEF probe R^2 rises from 0.14 to 0.48 (right). The forecast loss alone is insufficient; the uniformity term is the load-bearing ingredient.

Table 3. Phase decodability from EchoNet-Dynamic ED/ES keypoints (n_{pairs}=297). Closed loops alone (frozen) do not yield a linearly separable phase; fine-tuning the encoder does.

Variant	ED-vs-ES acc. ↑	shuffled	angular sep. (°)
Frozen encoder (d=128)	0.42	0.48	19.2
Frozen encoder (d=512)	0.42	0.49	18.7
λ_u=0 (collapse)	0.49	0.51	6.7
Shuffled-time control	0.41	0.50	17.0
Fine-tuned encoder	**0.66**	0.51	**12.6**

cleanly tracks geometry: ablating uniformity drops R^2 from 0.48 to 0.14, the same axis that drives $\widehat{d}_{\mathrm{ID}}$ and $\mathrm{LC}_{\mathrm{mean}}$ (Figure 1). Second, allowing the encoder to fine-tune on the same auxiliary objective delivers MAE 5.37, R^2=0.67, the best state-only EF probe we are aware of on EchoNet-Dynamic and a clean 0.15 R^2 gain over the raw baseline. Importantly the fine-tuned encoder retains $\widehat{d}_{\mathrm{ID}} \approx 2.4$ — the gain comes from *reparameterizing* the manifold, not from inflating its dimension.

4.5 Forecasting at clip stride: the persistence baseline is the right yardstick

A natural question is whether the transition operator f_θ itself, trained alongside the state head, is useful as a one-step forecaster. We compare f_θ to a persistence baseline ($\hat{\mathbf{z}}_{t+h} = \mathbf{z}_t$) at horizons $h \in \{1, \ldots, 8\}$. Across all 21 sweep cells, $L_1[f_\theta] > L_1[\text{persistence}]$ at every horizon (Table 5). The reason is geometric: on closed-loop

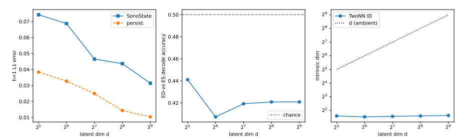

Fig. 2. Ambient-d sweep (S1). Intrinsic dimension and loop closure are invariant to head capacity; the state head is a probe, not a bottleneck.

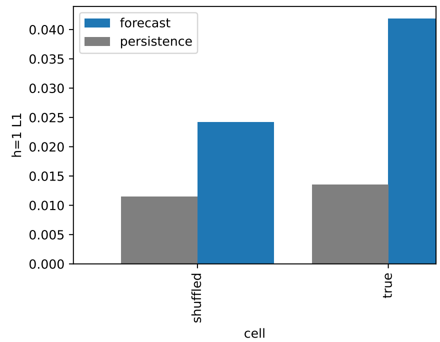

Fig. 3. C1 shuffled-time control. Loop closure is preserved by the on-sphere uniformity term alone, but intrinsic dimension rises slightly: true temporal causality is needed for the smooth periodic structure of the default cell.

trajectories with clip stride $\Delta t \approx 1\,$s on $\sim 2\,$s clips, adjacent latent states are nearly identical ($1{-}\cos \approx 0.04$), so persistence is an extremely strong baseline. The trained f_θ produces an order-of-magnitude larger cosine displacement (≈ 0.36 at $h{=}1$) and is therefore worse on absolute error.

This is a finding about *clip stride*, not about the state space. The two regimes where forecast and persistence coincide — $\lambda_u{=}0$ (collapse) and NORESIDUAL transition init (collapse) — are the same regimes where the geometric metrics flag a degenerate manifold (§4.2). When the manifold is well-formed, f_θ moves; when it is collapsed, f_θ is trivially good at persistence. A genuine forecasting evaluation in this setting needs either beat-to-beat strides (so adjacent states differ meaningfully) or intra-clip targets; we leave both to future work (§5).

4.6 Cycle-consistency as a complementary collapse detector

We fit a post-hoc inverse transition g_ϕ on a holdout set and report the round-trip error $\mathrm{RT}_h = \|g_\phi^h \circ f_\theta^h(\mathbf{z}) - \mathbf{z}\|_1$. On $n{=}353$ pairs the default cell gives $\mathrm{RT}_{1,2,4,8} = 0.038, 0.043, 0.049, 0.061$. The two ablations that visibly collapse the manifold ($\lambda_u{=}0$ and NORESIDUAL) also drive RT_1 to $\sim 10^{-4}$ — a counter-intuitive "best" result that, read alongside $\widehat{d}_{\mathrm{ID}}$, becomes diagnostic: very low RT_1 paired with very high $\widehat{d}_{\mathrm{ID}}$ indicates collapse, not quality. We therefore recommend reporting RT_h jointly with $\widehat{d}_{\mathrm{ID}}$ when probing the geometry of any latent world model.

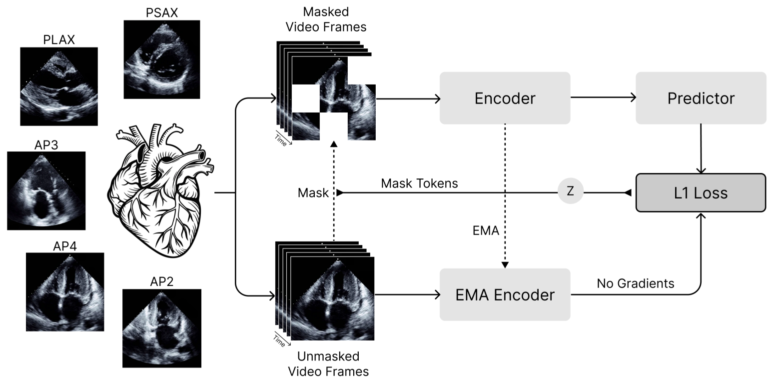

Fig. 4. Latent trajectories (qualitative). Held-out EchoNet-Dynamic videos projected into the first two PCs of $\mathbf{z}_t$. Each closed path is one video; closed loops emerge without any ECG or phase supervision.

5 Discussion

What we did and did not show. We showed that a small auxiliary objective compresses the 1024-dimensional V-JEPA-2-Echo feature space onto an intrinsically 3-dimensional closed manifold, and that the manifold's compactness (measured by $\widehat{d}_{\mathrm{ID}}$ and $\mathrm{LC}_{\mathrm{mean}}$) predicts downstream EF probe quality. We did *not* show a forecasting win over persistence at the chosen clip stride (§4.5), and we did *not* show that frozen-encoder states are linearly phase-decodable from a single clip (§4.3). Both negative findings point at concrete follow-ups (beat-to-beat training; encoder fine-tuning) rather than limitations of the geometric probe.

A geometric evaluation suite, demonstrated on V-JEPA-2. The three metrics we report ($\widehat{d}_{\mathrm{ID}}$, $\mathrm{LC}_{\mathrm{mean}}$, RT_h) are cheap to compute, do not require any phase or ECG labels, and produce a clear ablation signature on the V-JEPA-2 / EchoNet-Dynamic combination studied here. Whether the same signature transfers to other backbones (CNN-based, contrastive video-language) is an open empirical question that this suite makes it cheap to ask.

From a geometric probe to a conditioned world model. A faithful, low-dimensional state space is a precondition for, not a substitute for, a true world model. The natural extension is to introduce *interventions* $\mathbf{a}_t$ (a pharmacological step, a paced beat, a mechanical support change) and learn $f_\theta(\mathbf{z}_t, \mathbf{a}_t)$. The 3-D manifold we report here makes that extension tractable: counterfactual rollouts can be visualized in PCA, deviations from the unperturbed loop can be quantified geometrically, and a small state head is the natural conditioning point.

Limitations. (i) Our $\widehat{d}_{\mathrm{ID}} \approx 3$ is measured by TwoNN, a local estimator; we have not yet cross-checked it with persistent-homology or correlation-dimension estimators.

Table 4. LVEF regression on EchoNet-Dynamic (state-only ridge probe). The raw V-JEPA-2 mean-pool baseline (d=1024) beats the frozen state-only probe at small head dimension and is matched only at $d \geq 256$: the state head is a *geometric probe*, not a downstream-feature compressor. Encoder fine-tuning on the same auxiliary objective converts the geometric structure into a clean downstream win.

Backbone	MAE (%) ↓	R^2 ↑
Raw V-JEPA-2 mean-pool (d=1024, frozen)	6.52	0.52
SonoState (frozen, d=64)	7.71	0.35
SonoState (frozen, d=128)	7.55	0.38
SonoState (frozen, d=256)	7.27	0.43
SonoState (frozen, d=512)	6.91	0.47
SonoState (no uniformity, λ_u=0)	8.10	0.14
SonoState (sweet-spot, λ_u=10^{-2})	7.00	**0.48**
SonoState (fine-tuned encoder)	**5.37**	**0.67**

Table 5. Latent forecasting on EchoNet-Dynamic test split. Persistence is the dominant baseline at the clip stride used here because adjacent latent states are highly similar; see text. Lower is better.

	h=1		h=4	
Method	L_1 ↓	$1-\cos$ ↓	L_1 ↓	$1-\cos$ ↓
Persistence	0.025	0.067	0.025	0.073
SonoState (frozen enc., d=128)	0.046	0.219	0.066	0.376
SonoState (fine-tuned enc.)	0.042	0.360	0.044	0.395
SonoState (collapse, λ_u=0)	0.0003	0.00002	—	—

(ii) Loop closure is a within-video statistic; we have not verified that ED frames lie at consistent angular positions *across* patients (the phase-decodability negative finding suggests they do not, for frozen encoders). (iii) The 21-cell sweep is on a single dataset (EchoNet-Dynamic) and a single backbone (V-JEPA-2 ViT-L/16). Generalization to other echo FMs (EchoCLIP, CNN backbones) is open. (iv) Forecast evaluation at clip stride is structurally weak; beat-to-beat retraining is the obvious next experiment.

6 Conclusion

We presented a small auxiliary attachment for echocardiographic video foundation models that turns a 1024-dimensional V-JEPA-2 feature space into a 3-dimensional closed manifold whose geometry tracks the cardiac cycle. A 21-cell ablation isolates the on-sphere uniformity term as the single load-bearing ingredient: removing it inflates intrinsic dimension to 6, drops loop closure from 0.75 to 0.55, and slashes downstream EF probe R^2 from 0.48 to 0.14. Fine-tuning the encoder on the same objective yields a state-only EF probe at MAE 5.37,

R^2=0.67. We position the resulting three-metric probe (intrinsic dimension, loop closure, round-trip error) as a cheap, label-free geometric evaluation recipe that this paper instantiates on V-JEPA-2 / EchoNet-Dynamic and that lays the groundwork for future intervention-conditioned medical world models.

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